Exercise - 14

Implement Packet sniffing using RAW Sockets

Aim:

To capture and displays network packet using RAW sockets.

Algorithm:

1. Create a RAW socket
2. Receive packets continously.
3. Extract source MAC, destination MAC, and protocol
4. Identify IPV4, ARP or IPV6 packets
5. Display IP address and TTL for IPV4 packets
6. Stop using ctrl+c and close the socket.

Program:

```
import socket
import struct
s = socket.socket(socket.AF_PACKET,
    socket.SOCK_RAW, socket.ntohs(3))
print("Packet sniffer started...")
try:
    while True:
        packet, addr = s.recvfrom(65535)
dest, src, proto = struct.unpack
        ('!6s6sH', packet[:14])
    src_mac = ":".join("%02x" % b
        for b in src)
```

sample output:

Packet sniffer started...

Source MAC : 3c:52:82:11:22:33
Destination MAC : ff:ff:ff:ff:ff:ff
Protocol : 0x806
Packet Type : ARP

Source MAC : 3c:52:82:11:22:33
Destination MAC : 84:16:69:44:55:66
protocol : 0x800
Source IP : 192.168.1.10
Destination IP : 192.168.1.1
TTL : 64

```
        socket.inet_ntoa(ip[8]))
        print("TTL:", ip[5])
    elif proto == 0x0806:
        print("Packet Type: ARP")
    elif proto == 0x86DD:
        print("Packet Type: IPv6")
    else:
        print("Packet Type: Others")
except KeyboardInterrupt:
    print("\nSniffing stopped.")
finally:
    s.close()
```

Result:

Thus, packet sniffing was implemented successfully using RAW socket to capture and display network packet information